

Harsh Gupta

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Summary

Data Scientist with an MS in Data Science and 3+ years of production experience building predictive models, statistical frameworks, and ML systems across FSI, operations, and academia. Delivered \$1M–\$3M in profit impact through pricing elasticity modeling at Deloitte, with applied experience in time-series forecasting, NLP, LLM-based decision systems, and computer vision research. Skilled across applied ML workflows in Python, R, Scikit-learn, TensorFlow, and PyTorch, with production deployment experience on AWS and Azure.

Education

Indiana University Bloomington

Master of Science in Data Science (GPA: 4.0/4.0)

Manipal University

B.Tech in Electronics and Communication, Minor in Data Science (GPA: 8.13/10)

Aug 2024 – May 2026

Bloomington, IN

Jul 2017 – May 2021

Manipal, India

Technical Skills

Programming: Python, R, SQL, PySpark

ML & Modeling: Scikit-learn, TensorFlow, PyTorch, XGBoost, Regression, Statistical Inference, Predictive Modeling, Clustering, Scenario Modeling, Time Series Forecasting, Feature Engineering, A/B Testing, Hypothesis Testing, Model Evaluation, NLP

Cloud & Data Platforms: AWS, Azure, Redshift, OracleDB, Postgres, Tableau, PowerBI

Generative AI: LangChain, LangGraph, RAG, Azure AI Studio, Hugging Face

Data Engineering: ETL Pipelines, AWS Glue, Airflow, Azure Data Factory, Spark

Experience

Indiana University Bloomington

Part-Time Data Scientist | Python, Scikit-learn, Time Series Forecasting, Azure

Oct 2024 – Present

Bloomington, IN

- Analyzed millions of dining transaction records to develop **time-series forecasting models** for **football prediction**, reducing average forecast error from roughly **2,000 swipes to ± 500 swipes per day**.
- Partnered with operations and finance stakeholders to translate model outputs into **actionable planning strategies** for food preparation and labor scheduling.
- Drove an estimated **\$2K–\$3K daily operational cost optimization** through reduced waste and overstaffing, enabled by more precise model-driven scheduling decisions.

Indiana University Bloomington Campus Auxiliaries

AI & Data Science Intern | Python, Scikit-learn, LLM, Azure ML, LangChain

Jun 2025 – Aug 2025

Bloomington, IN

- Developed an **ML/LLM decision support system** for parking violation appeals, combining **appeal classification**, policy-clause NLP, and rejection pattern analysis to generate **contextual soft decisions** and structured policy summaries for human reviewers.
- Reduced appeal decision time from **weeks to days** by providing adjudicators with structured policy context, rejection signals, and LLM-generated rationale – reducing manual case research required per appeal.

Deloitte – Data & AI Consultant

Data Scientist – Customer Strategy & Pricing | Python, R, ML, Tableau, AWS S3

Sep 2021 – Jul 2024

Bengaluru, India

- Developed a **linear mixed-effects regression model** to estimate **store-level price elasticity** across 5,000+ locations, increasing profit by **\$1M–\$3M** while limiting customer churn to **1%**.
- Translated model outputs into strategic recommendations through **scenario analysis**, executive-ready presentations, and story-driven Tableau dashboards, informing pricing roadmaps and influencing leadership decisions.
- Led the full analytics lifecycle independently – **hypothesis framing**, **feature engineering**, **model design**, **cross-validation**, and executive-ready business interpretation – driving a shift from intuition-based to **model-driven pricing**.

Data Engineer – FSI Domain | AWS Glue, PySpark, OracleDB, Postgres

- Built nightly batch ETL pipelines in AWS Glue & PySpark processing **12–15M records/batch** across a **Landing** → **Staging** → **ODS** architecture, ensuring timely, reliable data for downstream analytics and reporting.

Samsung Research Institute

Research & Development Intern | Python, OpenCV, Computer Vision

Jan 2021 – Jun 2021

Remote

- Developed a **computer vision model** for low-light image enhancement, applying image preprocessing, feature extraction, and signal processing techniques in Python and OpenCV to improve mobile photography quality at the pixel level.
- Co-authored a research paper on the findings; received Samsung's **Excellence Award** for innovation and technical contribution.